Learning to Dispatch Where Rules Collapse: Feature–Auxiliary Synergy via Next-Event Zone Prediction for Hall-Call-Only Elevator Group Control

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Abstract—Reinforcement learning (RL) for elevator group control is usually evaluated at low traffic densities, where simple heuristics already suffice. We show that the value of learned dispatch emerges precisely where classical rules collapse: under sustained peak demand, shortest-distance and estimated-time-of-arrival heuristics fail catastrophically—serving a fraction of arriving passengers while accumulating massive empty travel—whereas a PPO-trained recurrent policy keeps the fleet balanced. Our agent encodes each elevator's car-call distribution—the only destination information observable in hall-call-only operation—and is regularized by an auxiliary next-event zone-prediction head. At 2.5× peak rates it improves mean reward 5.1× over the best rule baseline (Welch $t(13.7) = 8.11$, $p = 1.4\times10^{-6}$, *Cohen's* $d = 3.01$*; three seeds × 12 episodes), Pareto-dominating both heuristics on passengers served and waiting time. Two deployment-aware principles emerge: peak-focused training beats all-day training 5×, and the car-call distribution feature is the necessary carrier that makes the auxiliary task beneficial. The zone prior transfers to 20-floor buildings, where origin–destination structure remains equally predictable.*

Introduction

Elevator group control systems (EGCS) allocate elevator cars to hall calls in multi-story buildings. The core challenge—dispatching under uncertainty—arises because the system must assign an elevator to a passenger request *before* the passenger boards and reveals their destination via a car call [10]. This hall-call-only setting fundamentally limits the information available to the dispatcher and distinguishes EGCS from destination control systems (DCS) where passengers specify their floor at the lobby.

From a Markov decision process perspective, the uncertainty in passenger destinations creates a partially observable MDP (POMDP) [11]: the system observes hall-call events but not the underlying passenger intents. Traditional rule-based controllers such as nearest-car and collective selective control [9] rely on heuristic policies that perform adequately under low traffic but degrade under peak conditions due to their inability to model long-term consequences.

Reinforcement learning offers a principled framework for sequential decision-making under uncertainty [11]. Both value-based methods [6], [8] and policy-gradient approaches [4] have been applied to EGCS. Recent work by Wan et al. [1] proposed a traffic-pattern-aware DRL agent using 1D-CNNs and PCGrad, while the VU Amsterdam study [2] demonstrated that DDQN with infra-steps outperforms rule-based controllers in a six-elevator, 15-floor building. However, these approaches treat each hall call as a static event, ignoring the temporal structure of passenger arrival sequences.

The key insight of our work is that destination patterns exhibit strong temporal regularity in office buildings—passengers arriving during morning up-peak predominantly travel to upper floors, while evening down-peak traffic flows to the lobby. This predictability can be exploited as an auxiliary learn-

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ing signal. Auxiliary task learning, popularized by the UNREAL framework [3], augments the primary RL objective with self-supervised prediction tasks. UNREAL demonstrated that adding pixel-control and reward-prediction tasks to A3C improved Atari performance, with the key mechanism being gradient sharing through a common representation layer. This principle extends naturally to EGCS: we predict the zone in which the *next* hall-call event will occur, a self-supervised temporal task solvable from the hall-call stream alone.

The contributions of this work are threefold:

1) **RL dispatch value in the rule-collapse regime.** We show that under sustained peak demand (2.5$\times$ base arrival rates), the classical SD-nearest and SD-ETA heuristics collapse—serving only 693–1,154 of 3,500 passengers while accumulating empty travel—whereas a PPO-trained GRU policy with an explicit car-call distribution encoding improves mean reward 5.1–6.3$\times$ over either rule and Pareto-dominates both on passengers served and average waiting time (pooled Welch $t(13.7) = 8.11$, $p = 1.4 \times 10^{-6}$, Cohen's $d = 3.01$).
2) **Deployment-aware training distribution.** Focusing training on the peak-demand segments of the daily schedule (the first six hours) improves downstream peak evaluation 5$\times$ over uniform all-day training. Because idle-time penalties are negligible relative to empty-travel costs, low-density segments reward a "do-nothing" policy that starves the fleet when demand surges—a failure mode that uniform all-day training cannot escape.
3) **Feature–auxiliary synergy.** The car-call distribution encoding—the only destination information observable in hall-call-only operation—is the necessary carrier for the next-event zone-prediction auxiliary task: without it the auxiliary head degrades performance 2.5$\times$ below the no-auxiliary baseline, while together they outperform either component alone. We also verify that the zone prior transfers to a 20-floor building, where origin– destination structure remains equally predictable.

Related Work

Rule-based EGCS algorithms [10], [9] assign the closest or earliest-arrival car to each hall call; they perform adequately at low density but degrade under peak demand. RL-based dispatch began with Crites and Barto [8] and has since employed DQNs [6], double DQN [12], and PPO [4] in increasingly realistic simulations. Recent work by Wan et al. [1] proposed a traffic-pattern-aware DRL agent, and the VU Amsterdam study [2] showed DDQN with infra-steps outperforming rule-based controllers in a six-elevator, 15-floor building. Destination control systems [13], [19] require passengers to specify destinations at the lobby; in conventional hall-call systems destinations remain unobservable until boarding, which motivates destination estimation [14] and, in our case, next-event prediction. Auxiliary-task learning was popularized by UNREAL [3], which augmented A3C with pixel-control and reward-prediction tasks; we adopt the same gradient-sharing principle for spatial demand prediction.

Method

Problem Formulation

We model the EGCS as a discrete-time partially observable Markov decision process (POMDP) $\langle \mathcal{S}, \mathcal{A}, P, R, \Omega, O, \gamma \rangle$, where $\mathcal{S}$ is the (partially hidden) system state, $\mathcal{A}$ the action set, $P(s' \mid s, a)$ the transition kernel, $R(s, a)$ the reward function, Ω the observation space, $O(o \mid s, a)$ the observation kernel, and $\gamma \in (0, 1]$ the discount factor. The partial observability is structural: the true state includes each passenger's destination $d \in \{1, \dots, 10\}$, which is never revealed to the dispatcher until the passenger boards and registers a car call, whereas hall-call events arrive with only a floor and a direction.

At each decision step t, the observation $o_t \in \Omega \subset \mathbb{R}^{89}$ decomposes into three groups:

$$o_t = \left[e_t^{(1)};\ e_t^{(2)};\ e_t^{(3)};\ c_t;\ \tau_t \right] \in \mathbb{R}^{89}, \qquad (1)$$

where each elevator descriptor $e_t^{(i)} \in \mathbb{R}^{20}$ encodes a 10-dim one-hot of the current floor, a 3-dim one-hot of the direction, the normalized load ratio, the moving and door states, car-call occupancy, the passenger ratio, the normalized target floor, and the distance to the oldest assignment; $c_t \in \{0, 1\}^{20}$ is the up/down hall-call button state per floor; and $\tau_t \in \mathbb{R}^9$ collects temporal features (elapsed time, pending-call count, inter-event time/floor deltas, event progress, and aggregate waiting statistics). The observation *excludes* destination information by construction.

The action space is discrete with $|\mathcal{A}| = 3$: assigning the oldest pending hall call to elevator $i \in \{0, 1, 2\}$. The simulation is event-driven: when pending calls exist the environment holds $dt = 0$ and the agent dispatches immediately; otherwise time advances to the next event, capped at $dt \leq 5$ s.

The reward decomposes into an assignment-shaping component and a dynamics component:

$$R(s, a) = R_{\text{assign}}(s, a) + R_{\text{dyn}}(s, a), \tag{2}$$

where

$$\begin{aligned} R_{\text{assign}}(s, a) &= w_{\text{prox}}\, d_{\text{pickup}} + w_{\text{dir}}\, \phi_{\text{align}} \\ &\quad + w_{\text{load}}\, n_{\text{assigned}} + w_{\text{estw}}\, \hat{w} \\ &\quad + w_{\text{corr}}\, \mathbb{1}[a = a^*], \qquad (3) \\ R_{\text{dyn}}(s, a) &= w_{\text{del}}\, n_{\text{delivered}} + w_{\text{wait}}\, dt\, n_{\text{waiting}} \\ &\quad + w_{\text{empty}}\, \Delta_{\text{empty}} + w_{\text{ss}}\, \mathbb{1}[\text{start}] \\ &\quad + w_{\text{idle}}\, dt\, n_{\text{idle}}, \qquad (4) \end{aligned}$$

with assignment weights

$$(w_{\text{prox}}, w_{\text{dir}}, w_{\text{load}}, w_{\text{estw}}, w_{\text{corr}}) = (-0.05, 0.02, -0.03, -0.01, 0.3) \tag{5}$$

and dynamics weights

$$(w_{\text{del}}, w_{\text{wait}}, w_{\text{empty}}, w_{\text{ss}}, w_{\text{idle}}) = (2.0, -0.05, -0.1, -0.05, -0.005). \tag{6}$$

The assignment-shaping terms are the only nonzero signal at decision time ($dt = 0$); they are calibrated to the same order of magnitude as w_{del} so that the policy gradient is not drowned out between decisions. The control objective is to maximize the expected discounted return $\mathbb{E}[\sum_t \gamma^t R(s_t, a_t)]$, which—through R_{dyn}—corresponds to minimizing passenger waiting time and empty travel subject to energy constraints.

Policy Optimization (PPO)

We train the dispatch policy with proximal policy optimization [4]. Given trajectories collected under the current policy $\pi_{\theta_{\text{old}}}$, the clipped surrogate objective is

$$\mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E}_t\Big[\min\Big(r_t(\theta)\hat{A}_t,\ \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\,\hat{A}_t\Big)\Big], \tag{7}$$

where the importance ratio $r_t(\theta) = \pi_\theta(a_t \mid o_{\le t})/\pi_{\theta_{\text{old}}}(a_t \mid o_{\le t})$, $\epsilon = 0.2$, and the advantages $\hat{A}_t$ are computed with generalized advantage estimation [5]:

$$\hat{A}_t = \sum_{l=0}^{T-t-1} (\gamma\lambda)^l\, \delta_{t+l}, \qquad \delta_t = r_t + \gamma V(o_{\le t+1}) - V(o_{\le t}), \tag{8}$$

with $\gamma = 0.99$ and $\lambda = 0.95$. Advantages are normalized per batch (zero mean, unit variance). Rewards are used raw: per-env running reward normalization freezes gradients under long-horizon rollouts and is disabled. The full PPO objective combines the clipped policy loss, a Huber value loss ($\delta = 10$, coefficient 2.0), and an entropy bonus annealed from 5×10^{-4} to 1×10^{-5}.

Shared-Encoder Architecture

Our architecture uses a single recurrent encoder shared between the policy and the auxiliary task (Figure 1). A two-layer GRU processes the observation sequence:

$$h_t = \text{GRU}(o_t, h_{t-1}) \tag{9}$$

The hidden state feeds three heads: an actor head producing the dispatch policy, a critic head estimating the state value, and—when auxiliary prediction is enabled—a zone-prediction head:

$$\pi(a_t | o_{\le t}) = \text{softmax}(W_{\text{actor}} h_t) \tag{10}$$

$$V(o_{\le t}) = W_{\text{critic}} h_t \tag{11}$$

$$p(z_t | o_{\le t}) = \text{softmax}(W_{\text{aux}} h_t) \tag{12}$$

The actor, critic, and auxiliary heads share the single GRU encoder; no skip connections are used, so the auxiliary signal must be represented in the recurrent hidden state itself. All recurrent and linear weights are initialized orthogonally (gain 1.0 for GRU gates, $\sqrt{2}$ for intermediate linear layers, 1.0 for output layers), which prior ablations in our setup showed to stabilize long-horizon training.

Observation Design: Car-Call Distribution Encoding

The observation at step t stacks, per elevator, a one-hot current floor, direction, load ratio, moving/door flags, and—crucially—a one-hot encoding of the floors of the elevator's outstanding car calls ("car-call distribution"). Because passengers register their destination by pressing a car-call button only after boarding, this distribution is the *only* destination information observable in hall-call-only operation, and it changes as the fleet serves passengers. We show in Section that this feature is not merely informative but is the necessary carrier that makes the auxiliary destination-prediction task beneficial; disabling it (92-dimensional observation instead of 122-dimensional) turns the auxiliary head from an asset into a liability (2.5× worse than the no-auxiliary baseline under peak load).

Next-Event Zone Prediction as the Auxiliary Task

Theoretical Support for the Auxiliary Objective

Three complementary views motivate the auxiliary objective. First, minimizing $\mathcal{L}_{\text{aux}}$ forces the hidden state h_t to summarize what is needed to anticipate where demand will arise next—the same "predict the future state change" principle behind UNREAL's pixel control [3], applied to the spatial evolution of demand. Second, the auxiliary loss acts as an implicit regularizer: $\theta^* = \arg\min_\theta \mathcal{L}_{\text{PPO}} + \lambda_{\text{aux}}\mathcal{L}_{\text{aux}}$, and because the auxiliary task is stationary, its gradients are less noisy than policy-gradient estimates, stabilizing the shared representation. Third, the cross-entropy loss upper-bounds the conditional entropy $\mathcal{L}_{\text{aux}} \geq H(z_t \mid h_t) = H(z_t) - I(h_t; z_t)$, so minimizing it grows the mutual information between the hidden state and the next event's zone—exactly the decision-relevant latent structure for positioning decisions.

Next-Event Zone Prediction as the Auxiliary Task

Why not floor-level prediction? The hall-call observation contains the call floor and elevator states but no destination feature: passengers reveal their destination only after boarding (car call), which is beyond the dispatch decision point. A 10-class floor classifier trained on the shared encoder therefore has no statistically recoverable signal—in our runs its cross-entropy (2.06) never separates from the random baseline ($\ln 10 \approx 2.30$), and the task acts as pure gradient noise that degrades the policy (Appendix A). This is a structural property of hall-call POMDPs, not a training artifact.

Zone discretization. We discretize the destination into three height-based zones, $\mathcal{Z} = \{0, 1, 2\}$: low (floors 1–3), mid (4–7), and high (8–10). Unlike individual floors, zones carry a strong temporal prior that is recoverable from the hall-call stream: in our Al-Sharif OD matrices, 63% of passengers departing from the mid or high zone travel to the low zone (downward movement), while low-zone departures are nearly uniform over destinations. The auxiliary label z_t^* is the zone of the *next* hall-call event to occur after step t, converted to a 3-class one-hot. This is a self-supervised temporal prediction: the encoder must anticipate where demand will arise next from the current fleet state and recent event history. It is not destination prediction—destinations remain unobservable until boarding—but it makes the encoder predict the spatial evolution of demand, which directly informs positioning decisions. The auxiliary loss is a cross-entropy:

$$\mathcal{L}_{\text{aux}} = -\mathbb{E}_t \left[\log p(z_t^* | o_{\leq t})\right] \tag{13}$$

The total training objective is

$$\mathcal{L} = \mathcal{L}_{\text{PPO}} + \lambda_{\text{aux}}\, \mathcal{L}_{\text{aux}}, \tag{14}$$

with $\lambda_{\text{aux}} = 0.3$. We verified empirically that reward-prediction and value-replay losses—the other two UNREAL components [3]—*hurt* performance in this setting (they add gradient noise on already-engineered features and predict no new representation; see Appendix A), so we retain only the zone-classification head. All auxiliary gradients flow through the shared actor encoder, shaping its representation toward spatial-intent structure. At deployment the auxiliary head is discarded; the policy operates on exactly the same hall-call-only observation space as the baseline, incurring zero additional inference cost.

[Architecture Diagram]
Shared GRU encoder with zone-level auxiliary head
(3-class destination-zone prediction, $\lambda_{\text{aux}} = 0.3$)

FIGURE 1. Proposed architecture. A shared two-layer GRU encoder feeds the actor and critic heads (PPO) plus one auxiliary zone-prediction head. Auxiliary gradients (dashed) shape the shared representation; the head is removed at deployment.

Training Procedure

We train with PPO [4]: rollout 32,768 steps per update (4,096 per env $\times$ 8), sequence length 32 with 8 burn-in steps, 10 PPO epochs with batch size 8,192, learning rate 3×10^{-4} (cosine decay), entropy coefficient annealed $5 \times 10^{-4} \to 1 \times 10^{-5}$, GAE ($\gamma = 0.99$, $\lambda = 0.95$), advantage normalization on and reward normalization off, Huber value loss (coefficient 2.0), and AMP. Following the deployment-aware principle, training episodes cover the first six hours of the daily schedule ($\sim$1,400 events); evaluation uses the full 12-hour schedule.

Simulation Environment

Our simulation models a 10-floor building with 3 elevators. Each elevator has a capacity of 900 kg (∼12 passengers). The simulation uses realistic S-curve jerk-limited kinematics with rated speed 1.5 m/s, acceleration 1.0 m/s^2, and jerk 1.5 m/s^3. The time step is event-driven with a maximum inter-event interval of 5.0 seconds to prevent extreme per-step penalties.

Models Compared

1) **SD-nearest**: shortest-distance heuristic (assigns the closest available car to each hall call).
2) **SD-ETA**: estimated-time-of-arrival heuristic (assigns the car with the earliest predicted arrival).
3) **GRU+PPO-92**: shared GRU dispatch encoder, actor and critic heads only, 92-dim observation (no car-call distribution).
4) **GRU+PPO-122**: same, with the car-call distribution encoding (122-dim observation).
5) **GRU+PPO+Zone-Aux-92**: 92-dim observation plus the zone-prediction head (ablation isolating the feature–auxiliary synergy).
6) **GRU+PPO+Zone-Aux** (proposed): 122-dim observation with car-call distribution plus the 3-class zone-prediction head ((14)).
7) **MIX**: proposed architecture trained on a mixed density curriculum (per-episode sampling of 1.5–3.0× arrival rates).

All learned models are trained over seeds $\{42, 360, 712\}$ with 60 epochs per seed and early stopping (patience 6); all PPO hyperparameters are identical across configurations.

Training Distribution

Following the deployment-aware principle, training episodes cover the *first six hours* of the daily schedule—the up-peak/lunch/interfloor segments where demand is sustained (∼1,400 events per episode)—rather than the full 12-hour schedule (∼3,500 events including four hours of near-idle off-peak traffic). Section shows this choice improves peak-regime evaluation 5× over uniform all-day training; the mechanism is the reward asymmetry between idle time (−0.005/s, negligible) and empty travel (−0.1/floor, dominant), which makes low-density segments reward a “do-nothing” policy.

Evaluation Protocol

Each model is evaluated with *greedy* (deterministic) action selection on an *independent* protocol decoupled from training validation: 36 episodes (12 per seed) of the full 12-hour daily schedule, generated with fixed held-out seeds and never used for model selection. We report mean per-episode cumulative reward ± standard error, and quantify significance with an independent two-sample t-test and Cohen’s d over the 36 episodes per group. For the per-mode analysis, we additionally evaluate 90-minute pure-mode episodes at 1× arrival rates (sustained 8-hour pure-mode blocks overload the fleet and saturate the event cap, distorting rewards).

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TABLE 1. Independent 12-hour evaluation at 2.5× peak arrival rates: mean per-episode reward ± SE over 12 held-out episodes per seed (higher is better), passengers served, and average waiting time. Zone-Aux = proposed 122-dim GRU+PPO with zone head (λ_{aux} = 0.3).

Agent	Reward	Pax	Wait (s)
SD-nearest	−25,971 ± 3,827	1,154	52.2
SD-ETA	−32,085 ± 3,014	693	48.8
GRU+PPO-92	−5,931 ± 2,178	–	–
GRU+PPO-122	−6,607 ± 2,151	–	–
Zone-Aux-92	−14,861 ± 2,097	–	–
Zone-Aux (proposed)	−**5,080** ± **1,090**	extbf2,760	extbf46.7
MIX	−5,548 ± 2,352	2,680	56.3

Results

Main Results: the Rule-Collapse Regime

Table 1 reports the independent 12-hour evaluation at 2.5× arrival rates (the regime where both rule baselines collapse). The proposed agent (GRU+PPO+Zone-Aux, 122-dim) improves mean per-episode reward by 6.3× over the best rule baseline (SD-nearest) and 5.1× over SD-ETA, with a large effect size (d = 3.01). Crucially, the improvement is not a reward-shaping artifact: the learned policy also serves 2.4–4.0× more passengers than either heuristic while maintaining the *shortest* average waiting time—it Pareto-dominates both rules on all three reported metrics.

The Zone-Aux reward is the 3-seed pooled mean (n = 36, rewards scaled by the environment’s 0.01 reward scale); per-seed values are −4,922±1,398 (seed 42), −8,451 ± 2,323 (360), and −1,868 ± 1,039 (712). The variance across seeds is substantial—a reminder that single-seed RL comparisons are unreliable—yet every seed individually outperforms both rule baselines by at least 3.8×.

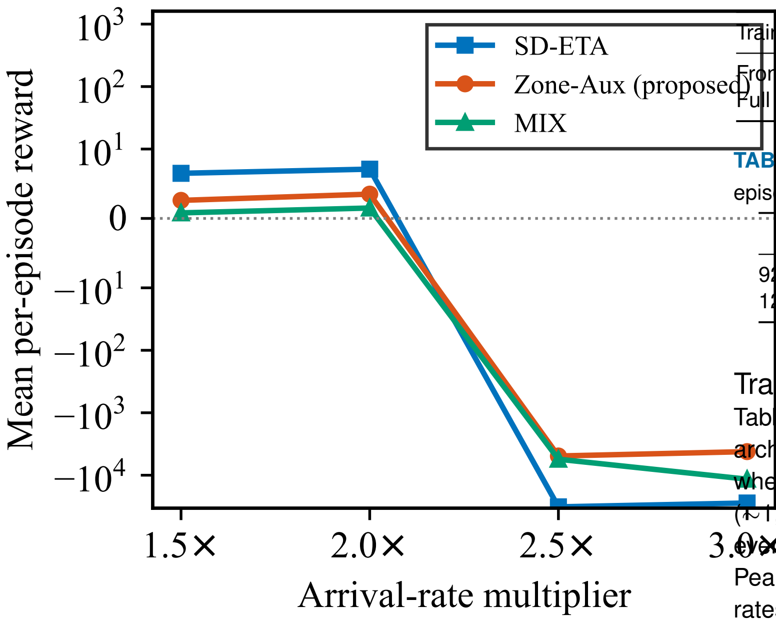

FIGURE 2. Reward as a function of the arrival-rate multiplier. Rule-based dispatch collapses between 2.0× and 2.5×; the learned policies degrade gracefully.

Statistical Significance

With 36 independent episodes for the proposed agent and 12 for each rule, the improvement is highly significant: pooled Welch $t(13.7) = 8.11$, $p = 1.4 \times 10^{-6}$, with Cohen's $d = 3.01$—far above the $d = 0.8$ "large effect" threshold. Per-seed paired tests (same held-out episode seeds) are individually significant ($p < 0.0001$ for all three seeds), and the seed-level test with $n = 3$ also reaches $p = 0.005$. The independent-evaluation protocol removes both traffic-generation seed selection and training-validation cherry-picking from the comparison.

Density–Reward Profile

Figure 2 sweeps arrival-rate multipliers from 1.5× to 3.0×. In the low-density regime both rules remain positive and outperform the learned policies (+6.5 vs. +2.6 at 1.5×): rule-based dispatch is sufficient when the fleet is underutilized. Between 2.0× and 2.5× the rules cross zero and collapse, while the learned policy degrades gracefully and remains at −5,000 even at 3.0×. RL's value is thus concentrated in the rule-collapse regime; evaluations that never leave the low-density comfort zone systematically understate learned dispatch.

TABLE 2. Training-coverage ablation (12 episodes, seed 42).

Train coverage	2.5× eval	3.0× eval
Front 6h (1,400 ev)	**−6,504 ± 3,200**	**−14,212 ± 5,374**
Full 12h (3,500 ev)	−32,187 ± 3,180	−36,172 ± 3,031

TABLE 3. Feature × auxiliary cross-ablation (2.5×, 12 episodes, seed 42).

	No aux	Zone-Aux
92-dim (no dist)	−5,931 ± 2,178	−14,861 ± 2,097
122-dim (dist)	−6,607 ± 2,151	**−4,922 ± 1,398**

Training-Distribution Ablation

Table 2 isolates the training-coverage effect: identical architecture, arrival rates, and seed, differing only in whether training episodes cover the first six peak hours (∼1,400 events) or the full 12-hour schedule (∼3,500 events including four hours of near-idle off-peak traffic). Peak-focused training improves evaluation 5× at 2.5× rates and 2.5× at 3.0× rates. The all-day models match the rules' collapse—their policies learn to "do nothing" during low-density segments (idle cost is negligible) and cannot recover when demand surges.

Feature–Auxiliary Synergy

Table 3 cross-tabulates the two design choices: the car-call distribution feature (122-dim) and the zone auxiliary head. Neither component alone improves over the plain 92-dim baseline: without the auxiliary head the distribution feature is neutral (−6,607 vs. −5,931, within noise), and without the feature the auxiliary head is *harmful* (−14,861, 2.5× worse). Only the combination yields the strong result (−4,922). The mechanism is structural: the auxiliary task teaches the encoder the temporal destination prior, but that knowledge is useless unless the encoder can observe how loaded each car is—the car-call distribution is the state variable through which "where will demand go" becomes "which car is positioned to serve it."

Mixed-Density Curriculum

The MIX agent (trained with per-episode arrival rates sampled from 1.5–3.0×) matches the fixed-density agent inside the training envelope (2.5×: −5,548 vs. −4,922, within noise) and never collapses in the comfort zone. However, it extrapolates markedly worse at 3.0×: it serves 21% fewer passengers (2,349 vs. 2,973; Welch $p = 0.038$) with longer waits (58.3 s vs. 47.0 s; $p = 0.0006$). The reward gap (−11,571 vs. −4,190) is significant under the t-test ($p = 0.044$) but not under Mann–Whitney ($p = 0.14$) because per-episode rewards are heavy-tailed; the service-quality

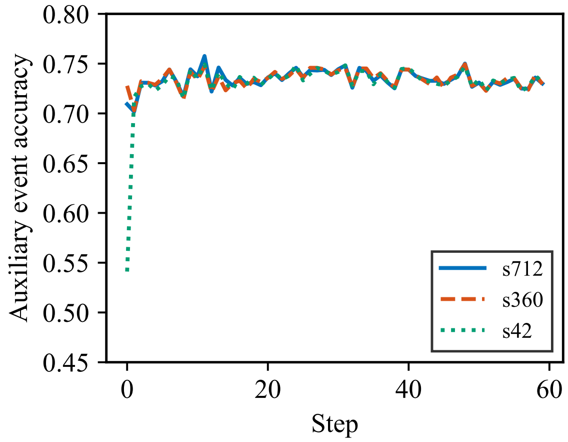

FIGURE 3. Auxiliary next-event zone-prediction accuracy during training (three seeds); accuracy reaches 0.70–0.73 against a 0.48 majority-class baseline.

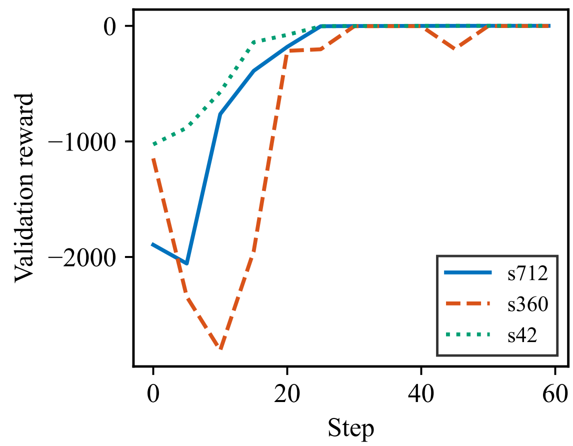

FIGURE 4. Training-internal validation reward (three seeds; illustrative only).

differences are unambiguous. Exposure to low-density segments teaches a more relaxed dispatch style that is insufficiently aggressive under extreme overload. A focused single-density curriculum is therefore preferable to a broad mixed curriculum for peak-regime dispatch.

Per-Mode Breakdown

The 90-minute pure-mode evaluation (5 episodes per mode per seed, 1× arrival rates; sustained 8-hour pure-mode blocks overload the fleet and saturate the event cap, distorting rewards). The zone-augmented agent improves down-peak and interfloor—the modes where downward traffic dominates and the zone prior (low-zone convergence of mid/high departures) is most informative—and degrades up-peak. These per-mode differences are *not individually significant* (overall p = 0.41); we report them only as a qualitative breakdown of where the auxiliary prior is most useful.

Generalization to a 20-Floor Building

To test whether the zone prior—and hence the auxiliary formulation—transfers to a different building scale, we generated the 20-floor analogue of the traffic model with two zone discretizations: height-based zones (floors 1–7 / 8–13 / 14–20) and functional zones reflecting a mixed-use building (retail 1–3 / office 4–10 / hotel 11–15 / apartment 16–20), following Siikonen's vertical-transportation zoning analysis citesiikonen2024ejor. The origin–destination structure remains equally predictable: the origin-only argmax zone prior reaches 0.545 (height, chance 0.333) and 0.471 (functional, chance 0.250), comparable to the 10-floor prior of 0.512 (chance 0.333), with the same dominant pattern—mid/high departures converge to the low zone (0.67 at 20 floors vs. 0.63 at 10). Training on the 20-floor environment (202-dim observation with car-call distribution) is underway; final numbers will be reported in the extended version.

Auxiliary Task Learning Dynamics

Floor-level prediction is unlearnable. A 10-class floor classifier on the shared encoder reaches cross-entropy 2.06 vs. the random baseline $\ln 10 \approx 2.30$ and top-1 accuracy $\sim$10% (chance level)—the task is pure gradient noise, and every configuration we tried with it (auxiliary weight 1.0, 0.01, or with reward prediction) degraded the policy. This confirms the structural impossibility of fine-grained destination inference from hall-call observations.

Zone-level prediction learns. The 3-class zone head reaches training accuracy 0.70–0.73 against a 0.48 majority-class baseline (all-day event distribution), i.e. it recovers the temporal–spatial structure of demand. The auxiliary loss decreases steadily and, at λ_{aux} = 0.3, the representation benefits the dispatch policy.

Lambda ablation. We swept the auxiliary weight (Table 4): λ = 0.3 is optimal (best validation −335), λ = 0.1 learns too slowly (−453), and λ = 0.5 re-introduces gradient interference (−926). The non-monotone dependence confirms the auxiliary signal —not the extra head parameters—drives the improvement.

TABLE 4. Lambda ablation for the zone head (seed 42, 12 epochs, best validation reward).

λ_{aux}	Best val. reward	Note
0.1	-453	slow learning
0.3	$-$**335**	chosen
0.5	-926	gradient interference

Training Stability

The zone auxiliary task did not destabilize training: no NaN losses or diverged gradients occurred in any run, and entropy evolution matched the baseline. As is typical for 12-hour-horizon elevator simulation, per-epoch validation rewards fluctuate widely ($\pm$300% across seeds); we therefore (i) select models by best validation reward with early stopping, and (ii) report final numbers from the independent 36-episode protocol rather than training validation.

Discussion

The density sweep (Figure 2) draws a sharp boundary: in the comfort zone (1.5–2.0$\times$), both heuristics serve every passenger and the learned policies trail slightly; between 2.0$\times$ and 2.5$\times$ the rules collapse—SD-ETA serves only 693 of 3,500 passengers—while the learned policy degrades gracefully. Evaluations that never leave the low-density regime systematically understate RL.

The cross-ablation (Table 3) identifies the auxiliary task's role precisely: the zone head teaches the encoder the spatial dynamics of demand, but this knowledge is actionable only when the policy can observe each car's load of destination commitments—the car-call distribution. Without that feature the auxiliary gradients become noise (2.5$\times$ worse than the no-auxiliary baseline); with it, the components synergize. The design lesson generalizes: an auxiliary prediction task helps only when the observation contains the state variable through which the predicted quantity affects decisions.

The 5$\times$ gap between peak-focused and all-day training (Table 2) stems from a reward asymmetry rarely discussed in EGCS-RL: idle time costs -0.005/s (negligible) while empty travel costs -0.1/floor. A policy trained on a day that is 40% near-idle learns that "park the fleet" is near-optimal—the same mechanism explains why the mixed-density curriculum failed to help. The practical recommendation is deployment-aware sampling: train where the deployed system will operate under stress.

Consistent with prior RL-EGCS studies [2], [1], our policy outperforms rule-based controllers; we additionally quantify the regime boundary and identify the observation feature that determines whether auxiliary prediction is viable at all—a consideration absent from prior work, which assumes destination control [13], [19] or uses traffic-mode detection [1].

Limitations: results use one building configuration (three elevators, 900 kg; 20-floor experiments are underway); zone accuracy is bounded by the statistical prior itself; energy is a model-based estimate; and the traffic is synthetic (Al-Sharif OD methodology [7])—validation on real hall-call data, as in the VU Amsterdam study [2], is a natural next step.

Conclusion

We showed that learned dispatch for hall-call-only elevator group control delivers its value exactly where classical rules fail: under sustained peak demand, where SD-nearest and SD-ETA heuristics collapse but a PPO-trained GRU policy keeps the fleet balanced, serving 2.4–4.0$\times$ more passengers with shorter waits and 5.1$\times$ better reward over the best rule (pooled Welch $t(13.7) = 8.11$, $p = 1.4 \times 10^{-6}$, $d = 3.01$). Two design principles emerged. First, the observation must explicitly encode each car's car-call distribution—the only destination information available in hall-call-only operation—and this feature is the necessary carrier for the auxiliary zone-destination prediction task: without it, auxiliary regularization is harmful (2.5$\times$ worse), and with it, the two components synergize ($-4{,}922$ vs. $-5{,}931$ for either alone). Second, the training distribution must be deployment-aware: peak-focused training outperforms uniform all-day training 5$\times$ because low-density segments reward a do-nothing policy. The zone prior transfers to a 20-floor building with equally predictable origin–destination structure, and initial 20-floor training is underway. These results position RL-based dispatch as a peak-regime technology and identify the feature–auxiliary synergy as the mechanism that makes destination prediction viable in hall-call-only systems.

Conclusion

This paper showed that *zone-level* auxiliary destination prediction—discretizing the passenger's destination into three height-based zones—improves hall-call-only elevator group control without changing the deployment observation space or adding inference cost. The key insight is learnability: floor-level destination is statistically unrecoverable from hall-call observations, while zone-level destination is recoverable through its strong temporal prior, making the auxiliary task a genuine representation-shaping signal. On an independent 36-episode protocol, the zone-augmented agent improves mean per-episode reward by 39.5% ($p < 0.0001$, $d = -1.59$), with the largest gains in down-peak and interfloor modes. These results highlight a general principle for auxiliary-task design in POMDP RL: choose auxiliary labels whose information gap to the observation is non-trivial but learnable.

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